

Technische Universität Hamburg (TU Hamburg)

M.Sc. Data Science

Module M1563: Research Project Computer Science

Passive Sensing Target Validation and Behavioral Forecasting in GLOBEM

Research Project Report (Studienarbeit)

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Abstract

This project examined what can be measured defensibly with longitudinal passive sensing in the GLOBEM dataset. The work used chronological evaluation, label-history baselines, construct-overlap checks, model-refitting controls, and explicit decision gates. It documented positive, negative, exploratory, and stopped analyses.

The strongest result concerns psychological-label validity. A prior same-person depression-label rate achieved held-out AUROC 0.844. Current-day passive features plus this baseline achieved 0.840, and passive history plus the baseline achieved 0.841. The passive features therefore showed no positive increment beyond label history in this comparison. Ten repeated psychological measures also overlapped substantially. The median absolute same-day correlation was 0.663 among the finite pairwise estimates. These results show that passive-state claims in GLOBEM require label-history and construct-overlap controls.

Directly observed behavior was easier to evaluate. Across 16 targets, pooled models using history, a 42-day fingerprint, and the current partial day achieved mean held-out $R^2 = 0.272$ and AUROC 0.788 for tomorrow. However, the proposed scalar “forecastability” score did not survive construct validation. Raw forecastability correlated $r = -0.912$ with behavioral standard deviation. After standard-deviation and marginal-entropy adjustment, split-half stability was $r = 0.230$. This failed the internally recorded 0.30 gate.

A narrower post-hoc result remained. Continuous profiles of relative history benefit were more similar within person-year trajectories than across matched trajectories after test-window, training-window, and sequence-proxy controls. Because both prediction models were pooled, this profile measures domain-specific transferability of a population-average history mapping. It does not establish an idiographic model, stable person-level phenotype, or clinical trait. Cross-domain coupling and discrete routine types were unsupported. A later personalization study stopped at an outcome-blind feasibility gate; no personalization model was run.

Data and Code Availability

Data. The report uses the de-identified GLOBEM v1.1 release described by Xu et al. [2022]. Restricted participant-level source data are not included in the submission repository. The repository contains code, analysis contracts, aggregate result files, checksums, and report figures. Source data must be obtained under the dataset’s access terms.

Repository. Code, contracts, public-safe aggregate outputs, and report sources are available at <https://github.com/Reaper2403/passive-sensing-target-validity>. The immutable release identifier for this report is `tuhh-project-report-v1`. The public repository was created from a clean, history-free snapshot so that restricted participant-derived artifacts in the private working repository are not exposed.

Software. Python scripts generated the CSV, JSON, and image outputs. Report figures were regenerated from aggregate outputs only. Tectonic is the sole tested document compiler. Exact source and output paths are listed in appendix A.

AI-assisted tools. OpenAI Codex was used as an assistive tool for document structuring, code drafting, and consistency checks. The author reviewed and verified the resulting code, analyses, citations, figures, and final text and remains responsible for the submitted work.

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Chapter 1

Project Scope and Objectives

1.1 Assigned problem

Passive sensing records behavior such as movement, sleep, screen interaction, and communication. Many studies use these records to predict depression, anxiety, or other psychological labels [Onnela and Rauch, 2016, Mohr et al., 2017, Saeb et al., 2015]. Prediction performance alone does not establish that the sensor data measure the named state. Performance can also arise from stable differences between people, persistence in repeated labels, related psychological constructs, or cohort-specific patterns.

The project task was to evaluate this measurement problem in GLOBEM and to identify a defensible target for passive-sensing analysis. The project therefore separated two questions:

1. Do passive features add information about repeated psychological labels after strong non-sensor baselines are included?
2. Can directly observed behavior be predicted prospectively from temporal history, and what does a stable prediction score represent after construct controls?

The work was not restricted to producing one positive model. It included baseline audits, robustness checks, negative tests, and stopping rules. This format matches the purpose of a research project report: the document explains the work performed, the decisions made, and the evidence supporting each conclusion.

1.2 Research questions

The final research questions were fixed for this report after the full project record and referee-driven audits were available.

RQ1. Psychological-label validity. Does passive sensing improve prospective depression-label classification beyond the person's prior label rate, and how separable are the repeated psychological outcomes?

RQ2. Behavioral forecasting. Can pooled models forecast directly observed behavior from chronological history across multiple sensing domains?

RQ3. Construct validity of forecastability. Is a scalar forecastability score stable after controlling behavioral variability and entropy-related proxies? If the scalar fails, does any continuous domain-profile result remain as a post-hoc lead?

RQ4. Mechanism and follow-up feasibility. Do cross-domain histories or discrete routine types explain the profile, and is the available data sufficient for a separately specified personalization study?

1.3 Project objectives and decision status

Table 1.1 states the objectives and their final status. “Passed” means that the recorded analysis satisfied its stated decision rule. “Failed” means the rule was evaluated and not met. “Post-hoc” marks analyses designed after inspection of earlier results. “Stopped” means an analysis ended at a feasibility gate before the intended model was fitted.

Table 1.1: Project objectives and final decision status. The status refers to the specific analysis in this report, not to a universal claim about passive sensing.

ID	Objective	Final status
O1	Audit psychological-label prediction against prior label history and construct overlap.	passed (boundary established)
O2	Demonstrate chronological forecasting of directly observed behaviors.	passed (measurement feasibility)
O3	Validate scalar forecastability after model-artifact and construct controls.	failed (variability-independent $r = 0.230 < 0.30$)
O4	Test a continuous domain profile after stronger controls.	post-hoc (internal lead only)
O5	Test cross-domain coupling and discrete behavioral types.	failed
O6	Run a frozen personalization comparison only if the data pass an outcome-blind feasibility gate.	stopped (no model run)

1.4 Deliverables and exclusions

The project deliverables are the analysis code, experiment definitions, aggregate outputs, claim-source map, reproducibility pipeline, and this report. The report is self-contained for the main conclusions, while the repository allows detailed inspection.

The report excludes the Amber product direction, Android application, volume-control study, and phone-based single-person observations. Those activities address product or pilot questions and do not support the GLOBEM research conclusions. The separate personalization feasibility audit is included only because it records a scientific stopping decision for a proposed follow-up.

Chapter 2

Background and Related Work

2.1 Passive sensing and digital phenotyping

Digital phenotyping uses data from personal digital devices to quantify behavior and health in daily life [Onnela and Rauch, 2016]. Personal sensing research combines ubiquitous sensors, repeated measurement, and statistical learning to study mental health outside the clinic [Mohr et al., 2017]. Common modalities include GPS, physical activity, sleep, screen use, app interaction, calls, Bluetooth, and Wi-Fi.

Early work reported associations between mobility or phone-use patterns and depressive symptoms [Saeb et al., 2015]. Later work has examined the timing of these associations. Stamatis et al. [2024], for example, separated concurrent and prospective utility and distinguished within-person from between-person effects. Balliu et al. [2024] evaluated person-specific mood prediction in a longer clinical design. These studies show that target definition, time scale, population, model unit, and validation split are part of the scientific claim.

Systematic reviews report substantial heterogeneity in sensors, preprocessing, outcomes, missing-data handling, and validation designs [De Angel et al., 2022]. This heterogeneity limits direct comparison of model scores. It also makes strong baselines important. A model should be compared with the information already available from prior labels, calendar structure, and behavioral history.

2.2 Generalization and target validity

GLOBEM was created to evaluate longitudinal human-behavior modeling across years and datasets. Its benchmark reported limited generalization for multiple depression detection methods [Xu et al., 2022]. This result does not by itself identify the cause. Possible causes include population shift, sensing differences, missingness, label construction, and model overfitting.

The present project focuses on target validity. A depression classifier can perform well because a person’s previous labels are persistent. It can also benefit from stable between-person prevalence. In that case, passive features may be predictive in isolation but provide little new information. A valid incremental test must compare passive sensing with a label-history baseline using a chronological split.

Repeated psychological scales create a second issue. Depression, anxiety, stress, and affect can share variance. Some released variables may also be transformations or subscales of the same instrument. A model score against one column therefore does not establish specificity

to the named construct. The project uses empirical correlation and temporal-prior models to quantify this overlap. It does not claim that overlapping constructs are clinically meaningless.

2.3 Behavioral predictability and regularity

Behavioral predictability is an established research topic. Song et al. [2010] used entropy-related methods to estimate limits on mobility predictability. Phillips et al. [2017] proposed the Sleep Regularity Index, which measures the probability of occupying the same sleep/wake state 24 hours apart. Wang et al. [2018] studied variability, circadian structure, and flexible regularity in smartphone sensing. Ren et al. [2022] used a continuous-time hidden Markov model to estimate activity rhythms.

These studies matter for construct validation. A prediction-error score can appear stable because a person has consistently low or high behavioral variance. A profile can also reflect marginal entropy, weekday structure, or short-term autocorrelation. The present project therefore treats variability and regularity as competing explanations, not as minor covariates.

The project does not estimate a theoretical predictability ceiling. Its entropy quantities are marginal or discretized descriptive proxies. Short trajectories and discretization prevent a direct Song-style interpretation. Phrases such as “learnable beyond entropy” are therefore not supported.

2.4 Between-person and idiographic interpretation

Stable smartphone behavior can predict broad between-person characteristics. Stachl et al. [2020] reported prediction of several Big Five traits from smartphone behavior. This literature provides a useful positive-control context: a pipeline can contain stable behavioral information even if it does not measure short-term psychological states.

Group-level models and person-specific models answer different questions. Group-to-individual generalization cannot be assumed [Fisher et al., 2018]. In this project, the forecasting coefficients were pooled across eligible training rows. Relative history benefit therefore measures how well a population-average history mapping transfers to a trajectory compared with a population-average calendar mapping. It is not an idiographic model.

Dense person-specific sampling is feasible in some settings. Across seven idiographic ecological momentary assessment studies, Soyster et al. [2019] reported high prompt completion. That result supports the feasibility of future longer person-specific studies, but it does not change the model unit in GLOBEM.

2.5 Validation design

Longitudinal data violate the independence assumption behind random row splits. Validation should respect temporal, hierarchical, and cohort structure [Roberts et al., 2017]. Small or structured samples can also produce uncertain cross-validation estimates [Varoquaux, 2018]. The project therefore used chronological splits, model-refitting checks, matched within-cohort permutations, bootstrap intervals, and leave-one-cohort-out tests.

The literature leads to four operational requirements used in this project:

1. compare passive psychological-label models with prior-label baselines;
2. forecast direct behavior prospectively rather than infer validity from cross-sectional association;

3. test prediction-derived constructs against variability and regularity; and
4. distinguish pooled-model transferability from person-specific learnability.

Chapter 3

Data and Reproducibility

3.1 GLOBEM dataset

GLOBEM v1.1 contains 705 person-years from 497 people across multiple longitudinal cohorts [Xu et al., 2022]. The present analyses used four institute-year cohorts from spring academic terms. Released RAPIDS features represented screen use, physical activity, sleep, location, calls, Bluetooth, and Wi-Fi.

The released identifiers do not provide a complete cross-year person key. The unit of analysis is therefore a *person-year trajectory*. It must not be interpreted as a unique person. This distinction affects all stability and profile statements.

Different stages of the pipeline contain different row counts because their eligibility rules differ. Table 3.1 keeps these stages separate.

Table 3.1: Analysis stages and units. Counts are not interchangeable because each stage applies different preprocessing and coverage rules.

Analysis stage	Recorded size	Unit and use
Published GLOBEM release	705 person-years; 497 people	Dataset-level description from Xu et al. [2022].
Label-baseline control	5,213 train rows; 2,927 test rows; 697 released identifiers	Chronological depression-label classification.
Frozen forecastability input	120,936 prediction rows; 696 trajectories; 16 targets	Prospective tomorrow predictions selected from the larger behavior forecast.
Original core split-half	521 trajectories	Eight-domain scalar and profile tests. Independently trained early and late models.
Disjoint-refit core split-half	499 trajectories	
Known-single-year sensitivity	232 trajectories	Conservative exclusion of all repeat-candidate trajectories from the fixed-model core sample.

3.2 Behavioral and psychological variables

The direct behavior forecast attempted 16 targets derived from released aggregate features. They covered screen interaction, physical activity, sleep, mobility, communication, Bluetooth, and Wi-Fi. Continuous targets were modeled on a $\log(1+y)$ scale. Binary targets were defined relative to the training-set median.

The core construct analysis used eight relatively common targets: screen-unlock count, screen-unlock duration, steps, active duration, sedentary duration, sleep duration, time in bed, and sleep efficiency. A 16-target rich-profile sensitivity was used only as a robustness check.

The psychological audit combined ten repeated outcome columns covering depression, anxiety, stress, and affect. There were 45 possible pairwise comparisons. Twenty-nine pairs had finite correlation estimates; the remaining pairs had no or insufficient same-day overlap. Several columns are derived or closely related representations, including PHQ-4 variants. The results therefore quantify empirical overlap in the released analysis table, not ten independent diagnostic instruments.

3.3 Preprocessing and missing data

The analysis used released daily aggregates rather than raw sensor streams. Each experiment applied its own minimum-history and coverage requirements. Missing data were not treated as proof of behavior or state. Coverage variables were included in selected controls, and the follow-up personalization audit used count-only eligibility checks before any outcomes were analyzed.

The behavior forecast used calendar features, yesterday’s outcome, rolling 7-, 14-, and 42-day summaries, and partial-day features where applicable. Features were constructed using information available before the prediction target. The target horizons were tomorrow and the following seven-day mean.

3.4 Cross-year identity boundary

Pinned GLOBEM overlap metadata listed 378 repeat-candidate trajectory keys. In the 521-trajectory core sample, 289 were candidates and 232 were known single-year trajectories. The public release did not permit reconstruction of a complete person cluster. A person-cluster bootstrap was therefore not possible.

The sensitivity analysis excluded all repeat candidates. It reduces the risk that a small set of known repeated users dominates the result. It does not prove stability within unique people across years. The report consequently uses “within-trajectory” and “person-year profile” rather than “stable person trait.”

3.5 Privacy and data access

This project is a secondary analysis of a de-identified research release. Recruitment, consent, and institutional-review procedures are described in the GLOBEM publication [Xu et al., 2022]. This report does not claim a new ethics approval. No raw GLOBEM data or participant-level follow-up tables are copied into the report folder or submission repository.

3.6 Repository and provenance

The repository contains only the experiments, aggregate results, and report sources used in this report. Machine-readable JSON and CSV files are authoritative for quantitative claims. The compact experiment record states analysis status and scope.

Each report figure is generated by `make_figures.py` from aggregate outputs. The repository file `report/CLAIM_SOURCE_MAP.md` maps every headline value and display to its checked output. The report folder contains no restricted input data.

Chapter 4

Project Method

4.1 Staged experimental design

The project proceeded in six stages:

1. establish passive-sensing and history baselines;
2. audit psychological labels against prior-label history and construct overlap;
3. forecast direct behavior prospectively;
4. define and test scalar and profile measures of forecast performance;
5. run construct, model-refitting, mechanism, and typology controls; and
6. freeze a separate personalization design and stop it if feasibility failed.

This order matters for interpretation. Early results motivated later checks. The later checks changed the status of the scalar forecastability claim. The report uses the final status rather than preserving an obsolete positive conclusion.

4.2 Chronological evaluation

For the main behavior forecast, each trajectory was split chronologically. The earlier 65% supplied the original training period, and the later 35% supplied prospective evaluation rows. No future target values were used to construct history features for an earlier prediction.

The psychological-label control also used a within-user temporal split. Its prior-label feature used expanding prior labels in training and the user’s training-period label rate for test rows. A training global rate was used when a user-specific rate was unavailable.

4.3 Psychological-label baselines

Random-forest depression classifiers compared five feature sets:

1. current-day all-day passive features;
2. 7/14-day passive history;
3. prior same-person depression-label rate only;

4. current-day passive features plus prior-label rate; and
5. passive history plus prior-label rate.

The primary metric was held-out AUROC. The incremental question was directional: did either combined model exceed the label-only model? A lower combined AUROC is reported as no positive increment, not as a small improvement.

For construct overlap, all possible same-day pairs among ten repeated psychological columns were examined. Ridge models then predicted each target using its own prior, other constructs' priors, or both. These models quantify overlap and persistence. They are not passive-sensing models.

4.4 Prospective behavior forecasting

Six model specifications were evaluated for each target, horizon, and issue time: calendar only; recent history; a 42-day fingerprint; partial-day behavior; history plus fingerprint; and history plus fingerprint plus partial-day behavior. Continuous targets used ridge regression. Binary high/low targets used logistic regression.

The models were pooled across eligible training rows. They were not fitted separately for each person-year trajectory. Mean held-out R^2 and AUROC were summarized across 16 heterogeneous targets. The means do not represent one task and have no common confidence interval in the recorded summary.

4.5 Model implementation

Table 4.1 records the settings needed to interpret the headline models. Settings are reported only when they are explicit in the runner or its configuration. The project configuration does not pin the scikit-learn version; therefore a setting left to a library default is identified rather than silently treated as an immutable parameter.

Table 4.1: Headline model implementation and explicit configuration. All scaling and imputation statistics were fitted on training rows.

Stage	Runner	Essential settings
Depression-label audit	<code>experiments/run_label_base_line_control.py</code>	Median imputation; random forest with 300 trees, minimum leaf size 5, balanced class weights, seed 42, and all CPU cores. No feature scaling.
Construct-prior audit	<code>experiments/run_construct_fragility_audit.py</code>	Training median imputation and standardization; closed-form ridge with $\alpha = 10$; unpenalized intercept; 65/35 within-trajectory temporal split.
Behavior forecast	<code>experiments/run_behavioural_weather_forecast.py</code>	Median imputation and standardization. Regression: ridge $\alpha = 1$. Classification: balanced logistic regression, <code>liblinear</code> , 1,000 iterations, seed 42; L2 penalty and $C = 1$ are scikit-learn defaults not overridden by the runner.
Forecastability refits	<code>experiments/run_referee_robusness_checks.py</code>	Custom median-imputed, standardized ridge with $\alpha = 1$; fixed, expanding, and disjoint temporal regimes; 1,000 bootstraps; 10,000 headline permutations; seed 240719.
Mechanism follow-up	<code>experiments/run_structure_mechanism_followup.py</code>	Same custom ridge with $\alpha = 1$; 1,000 bootstraps; 10,000 headline permutations; seed 211.
Typology follow-up	<code>experiments/run_profile_typology_followup.py</code>	Spherical k -means, $k = 2, \dots, 5$, 30 initializations and 100 iterations per fold, leave-one-cohort-out assignment, 500 permutations, seed 307.

4.6 Forecastability and relative history benefit

For trajectory u , target k , and held-out half h , absolute forecastability was defined as normalized negative mean absolute error:

$$F_{ukh} = -\frac{1}{\sigma_k} \frac{1}{|I_{ukh}|} \sum_{i \in I_{ukh}} |y_i - \hat{y}_i^{\text{hist}}|, \quad (4.1)$$

where σ_k is the target's global log-scale standard deviation. Higher values mean lower error. This score was the basis of the original scalar claim.

Relative history benefit was defined as

$$S_{ukh} = 1 - \frac{\text{MAE}_{ukh}^{\text{hist}}}{\text{MAE}_{ukh}^{\text{calendar}}}. \quad (4.2)$$

The numerator and denominator use pooled models. S measures where the pooled history mapping transfers better than the pooled calendar mapping. It is comparator-dependent and not a measure of separately fitted personal-model skill.

Target-half measures were standardized within target. Equal-weight averages formed scalar degree scores. Cross-target vectors formed continuous domain profiles. Profile similarity used cosine similarity.

4.7 Internally recorded gates

The original project record specified three gates before the full inferential run. The repository did not contain an independently timestamped prospective commit, so these gates are not described as preregistered.

- **H1 scalar stability:** early/late Pearson $r \geq 0.30$ with a trajectory-bootstrap 95% confidence interval above zero.
- **H2 profile stability:** same-trajectory early/late cosine similarity exceeds institute-year-matched cross-trajectory similarity by $\Delta \geq 0.10$ with permutation $p < 0.05$.
- **H3 trait association:** corrected association with selected traits has $|r| \geq 0.10$ and max-statistic permutation $p < 0.05$.

The 0.30 and 0.10 thresholds are project decision rules. They are not universal psychometric standards.

4.8 Model-artifact and construct controls

Three temporal model regimes tested whether shared fitted coefficients produced apparent stability:

Fixed. The original pooled model scored both test halves.

Expanding refit. The model was refitted after the first test half before scoring the final half.

Disjoint refit. Separate models were trained on disjoint early and late windows and evaluated on later windows.

Construct overlap was tested against observed target standard deviation and normalized marginal quintile entropy. A stronger sensitivity estimated these moments from the corresponding training block to avoid sharing the same held-out observations with MAE. Full-trajectory sequence controls added first-order conditional entropy, weekday-conditional entropy, and a Lempel–Ziv proxy. These are descriptive controls, not theoretical entropy-rate ceilings.

4.9 Uncertainty and multiplicity

Split-half correlations used 1,000 trajectory bootstrap samples. Matched profile contrasts used within-cohort permutations with finite-sample correction. Headline residual profiles used 10,000 permutations. Their minimum reported probability is $p = 0.0001$, rounded in prose as $p < 0.001$. Secondary feature-family and original profile tests used 500 permutations; their resolution floor is $p = 0.002$.

The H3 trait test used corrected inference. A later Fisher- z approximation estimated an 80% minimum detectable absolute correlation of about 0.172 under a conservative 28-test Bonferroni approximation. This was larger than the 0.10 effect gate, so a failed H3 is treated as inconclusive rather than evidence of independence.

4.10 Mechanism and typology follow-up

The post-hoc mechanism audit separated calendar terms, yesterday’s outcome, 7/14-day rolling means, a 42-day mean/standard-deviation fingerprint, all own-domain history, and recent histories from the other seven core domains. A feature family was considered an exploratory lead only if its adjusted scalar and profile gates held in all three model regimes.

Cross-domain coupling had an additional performance requirement: aggregate held-out MAE had to improve, and at least six of eight targets had to improve. Discrete typology used spherical k -means for $k = 2$ through 5. Clusters were trained on three cohorts and tested without refitting in the fourth. The typology gate required median adjusted Rand index (ARI) above 0.20, sufficient prevalence, and successful assignment stability in at least three held-out cohorts, including under disjoint refitting.

4.11 Personalization feasibility gate

A separate contract proposed a leave-one-institute-year-out personalization study. Phase A was outcome-blind and count-only. It required at least 30 eligible person-targets in every institute-year and at least 160 in total for selected target families. The contract required the study to stop before modeling if no allowed branch passed. This prevented a data-dependent relaxation after seeing prediction outcomes.

4.12 Implemented work packages

Table 4.2 summarizes the report analyses. The sequence moved from strong baselines to direct behavioral targets, then to construct controls and bounded follow-up tests. This kept model complexity secondary to target and construct validity.

Table 4.2: Implemented work packages and decisive outcomes. Detailed runner, contract, and output paths are retained in appendix A and the repository source map.

WP	Work performed	Decisive outcome	Status
1	Label-history and construct audit	Label-only AUROC 0.844 exceeded both combined models; repeated outcomes overlapped.	Primary boundary
2	Prospective behavior forecast	Direct behaviors were forecastable across 16 targets under pooled chronological models.	passed
3	Forecastability construct audit	Raw stability survived refitting but was largely variability; adjusted $r = 0.230 < 0.30$.	failed
4	Post-hoc structure and typology	Continuous profile survived tested controls; cross-domain increment and discrete typology failed.	post-hoc/mixed
5	Personalization feasibility	Outcome-blind eligibility left no allowed multi-family branch; no model was fitted.	stopped

The original forecastability run had passed its raw scalar and profile gates. The later construct audit materially changed that interpretation, so the final status in this report supersedes the earlier positive label. The personalization review, in contrast, accepted only the reproducibility of a stopping record; it did not assess a prediction effect.

Chapter 5

Results

5.1 RQ1: psychological-label validity boundary

5.1.1 Prior labels exceeded passive models

Current-day passive features achieved AUROC 0.579. Passive history achieved 0.691. The prior same-person label rate alone achieved 0.844. Current-day features plus the label rate achieved 0.840, and passive history plus the label rate achieved 0.841. Both combined models were lower than the label-only baseline (Figure 5.1 and Table 5.1). There was no positive increment beyond label history.

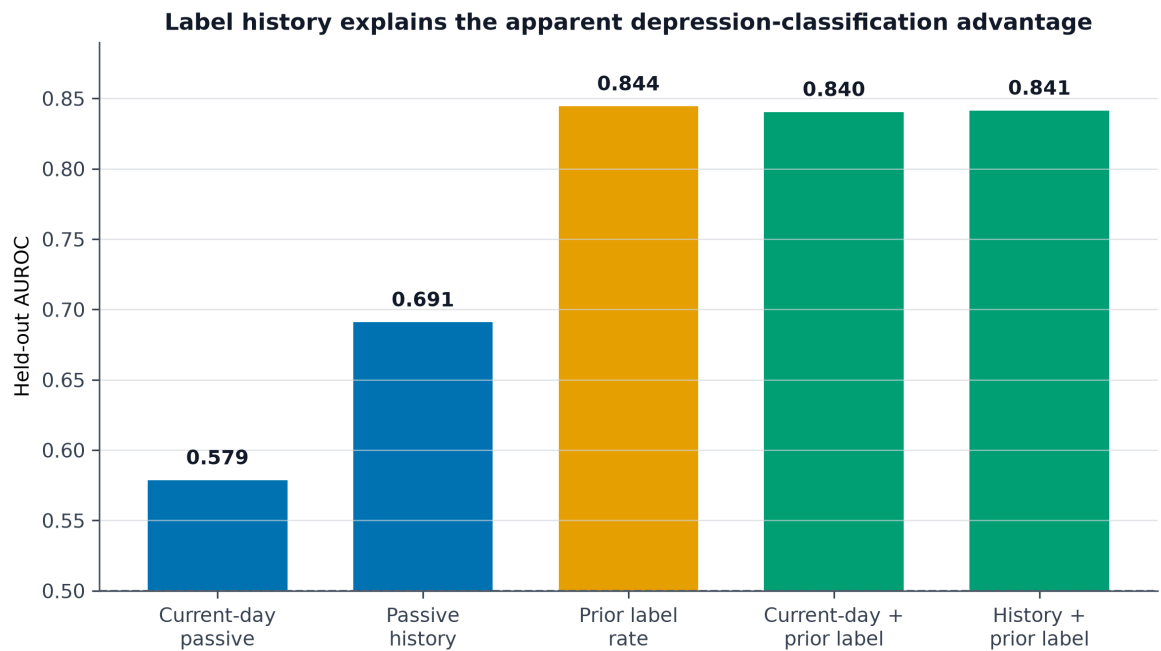


Figure 5.1: Held-out depression-label AUROC in the chronological label-baseline control. The analysis used 5,213 training rows and 2,927 test rows at this pipeline stage. The label-only baseline scored 0.844. Current-day plus label and history plus label scored 0.840 and 0.841, respectively, and did not improve on label-only. The prior-label feature is a self-report persistence/prevalence baseline, not a passive sensing model.

Table 5.1: Depression-label classification results. Differences are calculated against the unrounded label-only AUROC of 0.844428.

Feature set	AUROC	Difference vs. label-only	Interpretation
Current-day passive	0.579	−0.266	Passive-only
Passive history	0.691	−0.154	Passive-only
Prior label rate only	0.844	0.000	Strongest baseline
Current-day passive + prior label	0.840	−0.004	No positive increment
Passive history + prior label	0.841	−0.003	No positive increment

This comparison is cohort- and split-specific. It does not show that passive sensing can never add value. It shows that the tested passive features did not improve on a strong label-history baseline in this GLOBEM analysis.

5.1.2 Psychological measures overlapped

The construct audit contained 17,346 rows with at least one psychological measure. Among the 29 finite estimates from 45 possible same-day pairs, the median absolute Pearson correlation was 0.663. Figure 5.2 shows both the strong correlations and the missing pair estimates caused by absent or insufficient same-day overlap.

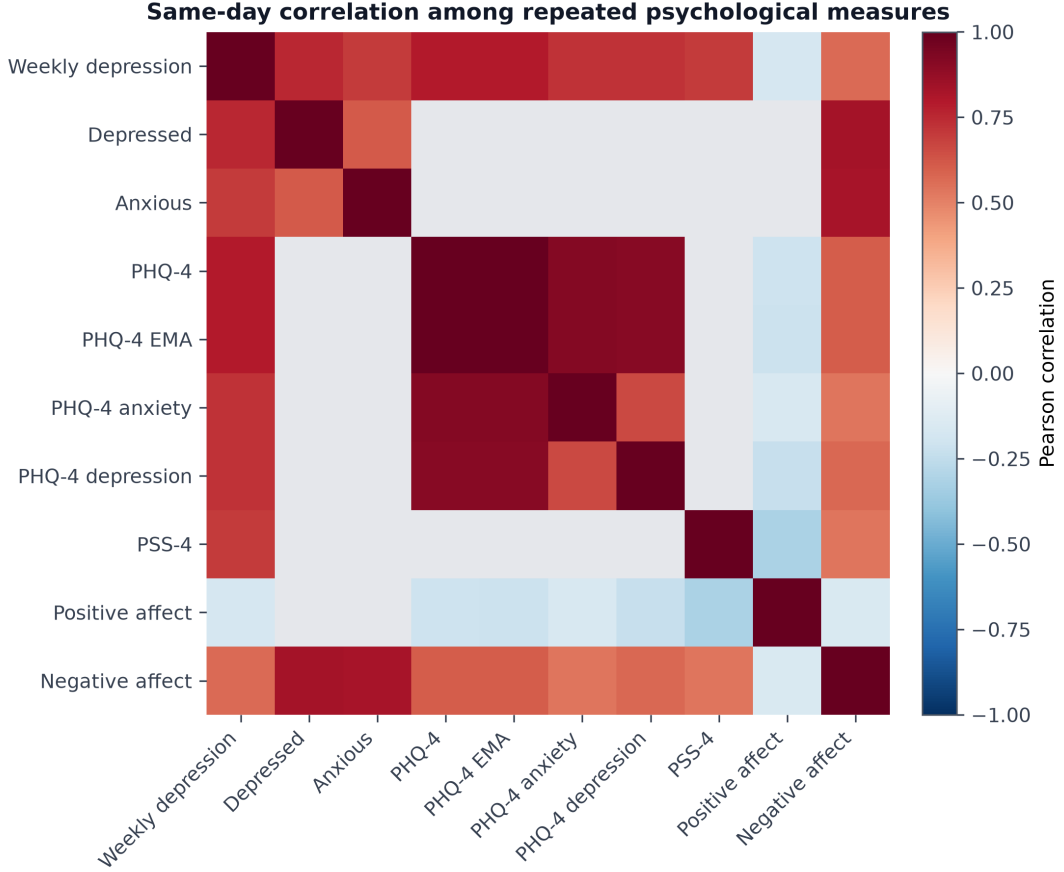


Figure 5.2: Same-day Pearson correlation among ten repeated psychological columns. Gray cells have no finite estimate because same-day overlap was absent or insufficient. The median absolute correlation was 0.663 among 29 finite pairwise estimates from 45 possible pairs. Several columns are derived or closely related PHQ-4 representations, so the figure documents empirical overlap rather than ten independent diagnostic constructs.

Mean held-out R^2 was 0.479 when each target used its own prior and 0.426 when it used only other constructs' priors. Adding other priors to the own-prior model increased mean R^2 by 0.044. Other-construct priors approached but did not equal own-prior prediction on average. No formal equivalence test was performed.

5.2 RQ2: directly observed behavior was forecastable

Across 16 targets, the full evening model achieved mean held-out $R^2 = 0.272$ and AUROC 0.788 for tomorrow. For the following seven-day aggregate, the corresponding means were $R^2 = 0.506$ and AUROC 0.871 (Figure 5.3 and Table 5.2). These are means across heterogeneous tasks and not the performance of one target.

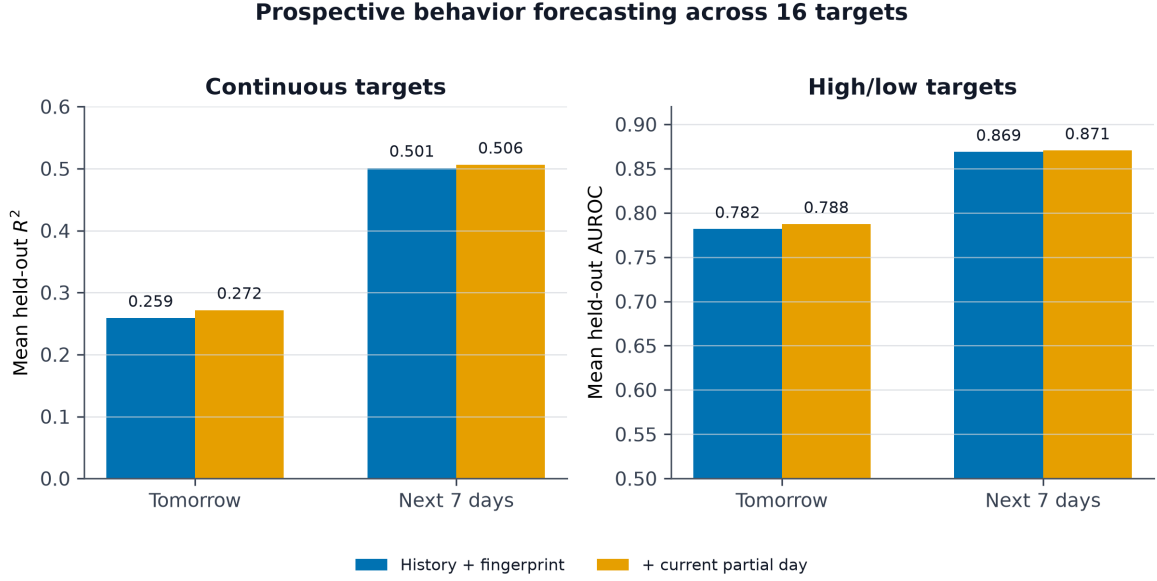


Figure 5.3: Prospective performance averaged across 16 behavioral targets for the evening issue time. Blue bars use pooled calendar, recent-history, and 42-day fingerprint terms. Orange bars also include current partial-day behavior. Continuous targets use held-out R^2 ; high/low targets use held-out AUROC. No common confidence interval is available for these target means.

Table 5.2: Aggregate behavior forecast at the evening issue time. “Lift” is the difference after adding the current partial day to history plus fingerprint.

Horizon	Metric	History + fingerprint	+ partial day	Lift
Tomorrow	R^2	0.259	0.272	0.013
Tomorrow	AUROC	0.782	0.788	0.006
Seven-day mean	R^2	0.501	0.506	0.005
Seven-day mean	AUROC	0.869	0.871	0.002

Most aggregate performance existed before adding the partial day. The result shows prospective continuity in observed behavior under pooled models. It does not establish psychological-state prediction, causal relationships, or personal-model learning.

5.3 RQ3: scalar forecastability failed construct validation

5.3.1 Raw stability survived model refitting

The original fixed-model scalar correlation was $r = 0.412$ with a bootstrap 95% interval of $[0.316, 0.501]$ for 521 trajectories. Expanding refitting produced $r = 0.407$ with interval $[0.307, 0.496]$ for 521 trajectories. Fully disjoint refitting produced $r = 0.322$ with interval $[0.219, 0.414]$ for 499 trajectories. Shared coefficients did not fully explain the raw association.

The raw domain-profile advantage also persisted: $\Delta = 0.287$ in the original fixed analysis ($p = 0.002$, 500 permutations), $\Delta = 0.280$ under expanding refitting, and $\Delta = 0.266$ under disjoint refitting (both $p < 0.001$, 10,000 permutations). These checks address model reuse. They do not establish construct validity.

5.3.2 Variability explained the scalar result

At the person-average composite level, raw absolute forecastability correlated $r = -0.912$ with observed behavioral standard deviation. The Spearman correlation was -0.909 . The score was therefore almost an inverse-variability measure under this implementation.

After standard-deviation and marginal-entropy adjustment, fixed-model split-half stability fell to $r = 0.230$ with interval $[0.049, 0.409]$. It did not meet the internally recorded 0.30 H1 gate. The corresponding expanding and disjoint values were 0.239 and 0.198. Figure 5.4 and Table 5.3 reports the complete comparison.

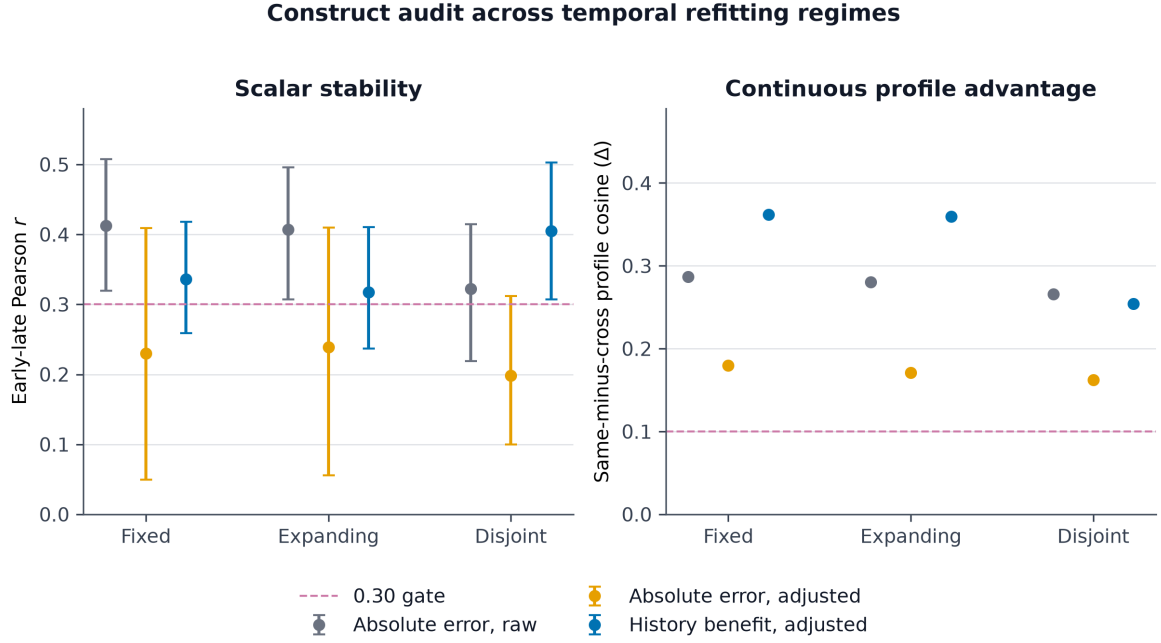


Figure 5.4: Construct audit for the core eight-domain sample. Scalar stability uses 521 trajectories in fixed and expanding regimes and 499 in the disjoint regime; error bars are trajectory-bootstrap 95% intervals. Profile advantage is the same-minus-matched-cross early/late cosine difference. The magenta lines show the internally recorded 0.30 scalar gate and the 0.10 profile gate. Adjusted rows control test-window target standard deviation and marginal quintile entropy. Profile tests used 10,000 permutations and had $p < 0.001$.

Table 5.3: Core construct-audit results. The adjusted absolute-error scalar fails the 0.30 gate in every regime. Relative history benefit is model-relative and remains subject to the stronger controls in Table 5.4. The fixed raw confidence interval is from the robustness rerun; the frozen original run reported $[0.316, 0.501]$.

Measure	Regime	Degree r	95% CI	Profile Δ	Status
Absolute error, raw	Fixed	0.412	$[0.319, 0.507]$	0.287	Raw gate met
	Expanding	0.407	$[0.307, 0.496]$	0.280	Raw gate met
	Disjoint	0.322	$[0.219, 0.414]$	0.266	Raw gate met
Absolute error, SD/entropy adjusted	Fixed	0.230	$[0.049, 0.409]$	0.180	Scalar failed
	Expanding	0.239	$[0.056, 0.409]$	0.171	Scalar failed
	Disjoint	0.198	$[0.100, 0.312]$	0.162	Scalar failed
History benefit, SD/entropy adjusted	Fixed	0.336	$[0.259, 0.418]$	0.362	post-hoc
	Expanding	0.317	$[0.237, 0.410]$	0.360	post-hoc
	Disjoint	0.405	$[0.307, 0.503]$	0.254	post-hoc

The correct conclusion is direct: the stable component of raw forecastability is largely stable behavioral variability. The positive residual correlation does not rescue the scalar claim because it failed its own threshold. The original scalar phenotype interpretation did not survive.

5.3.3 The continuous profile remained a post-hoc lead

Relative history benefit was less coupled to standard deviation than raw error at the person-average level ($r = -0.268$), but it still correlated with marginal entropy ($r = -0.518$). Its raw split-half stability was 0.484, 0.473, and 0.445 under fixed, expanding, and disjoint regimes.

The scalar did not survive every stronger control. Training-window moment adjustment produced $r = 0.323$, 0.263, and 0.322; the expanding value failed the 0.30 gate. Sequence-proxy adjustment produced $r = 0.240$, 0.219, and 0.246; all failed. The history-benefit scalar is therefore not a robust phenotype.

The continuous profile was more consistent. Same-minus-cross profile advantages remained above 0.10 after three types of adjustment (Table 5.4). All listed tests used 10,000 within-cohort permutations and had $p < 0.001$.

Table 5.4: Post-hoc continuous profile of pooled relative history benefit under stronger controls. Values are same-minus-matched-cross cosine differences. Sample sizes for sequence adjustment are 439, 439, and 425 trajectories for fixed, expanding, and disjoint regimes.

Adjustment	Fixed Δ	Expanding Δ	Disjoint Δ	Status
Test-window SD and marginal entropy	0.362	0.360	0.254	post-hoc
Training-window SD and marginal entropy	0.354	0.311	0.217	post-hoc
Sequence-order proxies	0.314	0.308	0.202	post-hoc

This is the surviving secondary result. It is a continuous within-trajectory profile of where pooled history features transfer relative to pooled calendar features. It has not been replicated in a second dataset. It is not evidence of idiographic learnability or a stable person-level trait.

5.4 RQ4: mechanism, typology, and feasibility

5.4.1 Own-domain multiscale history carried the profile

After test-window standard-deviation and marginal-entropy adjustment, 7/14-day means, their recent-history combination, the 42-day fingerprint, and all own-domain history passed the exploratory scalar and profile gates across the three regimes. Yesterday alone did not pass the scalar gate in every regime (Figure 5.5).

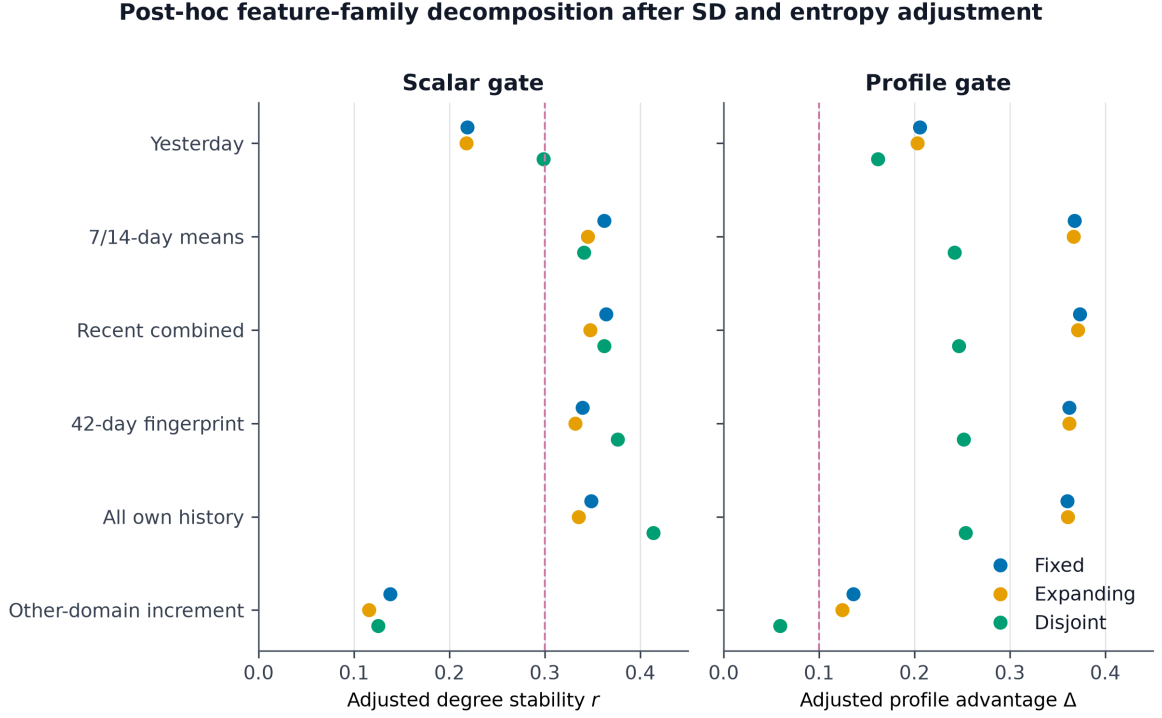


Figure 5.5: Post-hoc feature-family results after target-specific test-window standard deviation and marginal-entropy adjustment. Degree stability is early/late Pearson r ; profile advantage is the same-minus-matched-cross cosine difference. Magenta lines show exploratory 0.30 and 0.10 gates. Fixed, expanding, and disjoint core samples contain 521, 521, and 499 trajectories. Secondary profile tests used 500 permutations and reached the resolution floor $p \leq 0.002$.

The feature families overlap. Recent history and the 42-day fingerprint produced similar aggregate errors, and their combination yielded only a small additional change. The analysis does not identify one exclusive mechanism.

5.4.2 Cross-domain coupling was unsupported

Adding other-domain recent histories worsened aggregate held-out MAE by 0.47%, 0.44%, and 0.42% under fixed, expanding, and disjoint refitting. It improved four, four, and two of eight targets. The cross-domain performance gate required an aggregate improvement and at least six improved targets. It failed in every regime.

The surviving profile should therefore not be explained as cross-domain behavioral coupling. The supported description is multiscale own-domain pooled-model transferability under the tested controls.

5.4.3 No discrete routine types replicated

No k from 2 through 5 passed the leave-one-cohort-out typology gate (Figure 5.6). Median ARI ranged from 0.034 to 0.066 for fixed models and 0.028 to 0.057 for disjoint models. Every candidate had zero held-out cohorts that met the joint rule.

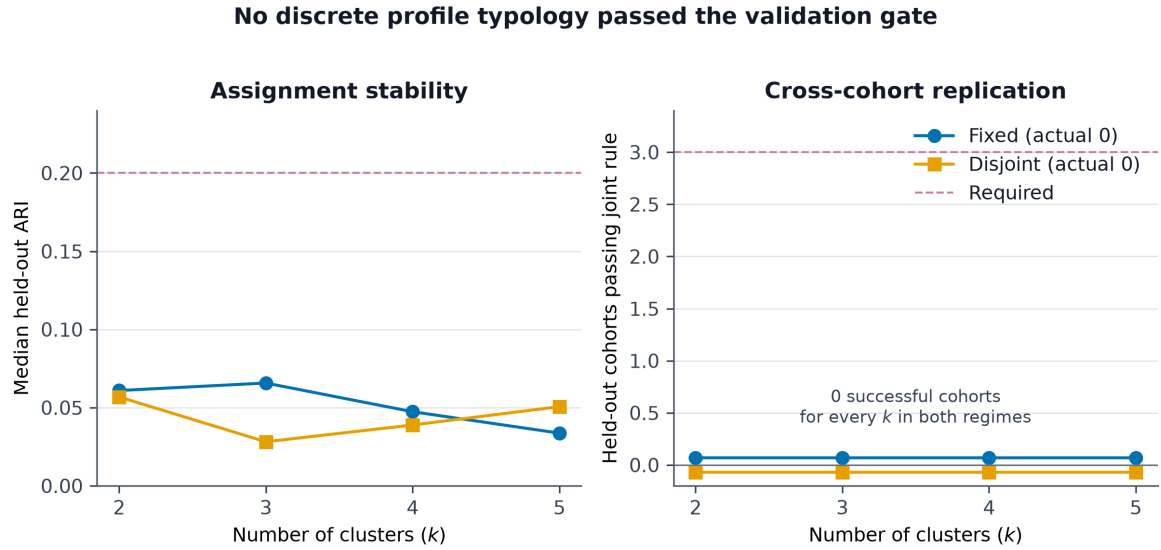


Figure 5.6: Post-hoc leave-one-cohort-out validation of spherical profile clustering. The left panel shows median adjusted Rand index across held-out cohorts; the right panel shows the number of held-out cohorts meeting the joint stability rule. Magenta lines mark the exploratory requirements of $\text{ARI} > 0.20$ and at least three successful cohorts. Both right-panel series equal zero for every k ; their markers are offset slightly only so both series remain visible. No $k = 2, \dots, 5$ passed in fixed or disjoint regimes.

Descriptive centroids can be given intuitive labels, but participants did not retain those assignments. The report therefore does not define sleep-, activity-, or screen-anchored behavioral types.

5.4.4 Personalization study stopped before modeling

The preferred branch of the outcome-blind feasibility audit retained only mobility and phone-interaction families. The selected mobility target had a minimum of 36 eligible person-targets across institute-years and 329 in total. The selected phone-interaction target also had a minimum of 36 and 327 in total. Physical activity and sleep had minima of 14 and 13, below the required 30. The reduced branch retained no family.

The contract required an allowed multi-family branch. None remained, so the selected branch and selected targets were empty. No model, prediction, loss, or outcome comparison was run. The correct status is **stopped** at a feasibility boundary, not a personalization null.

5.5 Secondary and exploratory findings

No corrected association with conscientiousness, extraversion, or social fit passed H3. The test had limited sensitivity for effects near the 0.10 gate, so the result is **inconclusive**. It does not establish independence from personality.

The common-distress follow-up was post-hoc and cross-sectional. The original controlled estimate was $r = -0.178$ for 518 trajectories. Direct target-standard-deviation control gave $r = -0.140$, with cohort values ranging from approximately 0.008 to -0.342 . Because the estimates were heterogeneous and unreplicated, they are excluded from the abstract and central conclusion.

Chapter 6

Discussion and Lessons Learned

6.1 Evaluation against the research questions

RQ1 established the project’s primary validity boundary: in this GLOBEM comparison, passive features did not improve the depression label beyond label history, and the repeated outcomes overlapped substantially. RQ2 passed as measurement feasibility: direct behavior was forecastable under pooled chronological models. RQ3 failed for the scalar construct because the adjusted value was $r = 0.230 < 0.30$; only a post-hoc continuous profile remained. RQ4 produced boundaries rather than a new mechanism: cross-domain history and discrete typology were unsupported, and the separate personalization design stopped before modeling. These answers are specific to the implemented data, splits, models, and decision rules.

6.2 Implication for passive-sensing target choice

The project supports a limited but useful conclusion. In this GLOBEM comparison, prior self-report history was more predictive of the depression label than the tested passive features. The repeated psychological columns also shared substantial variance. A passive classifier can therefore appear effective without adding new measurement of a named psychological state.

This result changes the evaluation standard. A current-day passive model should not be judged only against chance or another passive model. It should be judged against the person’s available label history. A multi-construct dataset should also examine whether the target is empirically distinct from related outcomes.

The result does not reject passive sensing for mental-health research. Passive features may support monitoring, context, adherence, or carefully designed within-person studies. The project shows what must be demonstrated before making a state-inference claim.

6.3 What forecastability measured

The scalar audit is the most important correction made during the project. A split-half correlation of 0.412 initially looked like a stable new construct. The -0.912 correlation with behavioral standard deviation showed that low error mostly identified low-variance behavior. The residual $r = 0.230$ failed the recorded gate.

This is not a failure of the forecasting pipeline. The raw association remained under expanding and disjoint refitting. The pipeline measured a reproducible quantity, but that quantity was mainly a known one: behavioral variability. The scientific task was construct validation, not only reproducibility.

Relative history benefit reduced the direct SD coupling, but it remained related to entropy and comparator behavior. Its scalar result also failed the stronger sequence controls. A ratio score is not automatically a valid trait measure.

6.4 Meaning of the continuous domain profile

The post-hoc profile result is narrower. It states that the pattern of domains in which a pooled history model outperformed a pooled calendar model was more similar within a person-year trajectory than across matched trajectories. This held after the tested moment and sequence controls.

Three boundaries are essential:

1. the models use population-average coefficients, so the profile measures transferability rather than personal-model learnability;
2. the unit is a person-year trajectory, not a cross-year identified person; and
3. the result was developed post-hoc in one dataset and requires prospective external replication.

Within these boundaries, the profile is a reasonable future research target. It is continuous rather than categorical. It appears to use multiscale own-domain history. The present analysis does not justify a clinical label or personality interpretation.

6.5 Negative results as design information

The project produced several useful negative results. Passive features did not add to the prior-label baseline. Cross-domain histories did not improve the behavior forecast. Discrete clusters did not replicate across cohorts. The personalization design did not have sufficient balanced data for its required multi-family comparison.

These results prevent unnecessary complexity. They support simple baselines, prospective evaluation, continuous profiles, and feasibility checks before model fitting. They also show why a report of the full project is more appropriate than a paper organized around one positive claim.

6.6 Project-execution lessons

6.6.1 Strong baselines should be implemented early

The label-history control materially changed the interpretation of earlier passive model scores. It should have been part of the first comparison. For longitudinal labels, prior-label and prevalence baselines are not optional diagnostics.

6.6.2 Construct audits should follow prediction tests

Prediction performance and temporal reproducibility do not establish that a new construct exists. Variance, entropy, missingness, and comparator behavior should be tested directly. A later audit is still valuable, but an earlier audit would reduce rework.

6.6.3 Stopping rules protect follow-up studies

The personalization audit checked feasibility without examining outcomes. Its stopping decision avoided fitting a model to an underpowered or unbalanced target set and then redefining

the question after seeing results.

6.6.4 Status labels reduce overclaiming

The repository distinguishes internally recorded tests, post-hoc analyses, exploratory controls, and analyses stopped before modeling. This distinction should remain in the final submission. A low p -value in a post-hoc analysis does not convert it into a confirmatory result.

6.7 Threats to validity

Population and season. GLOBEM contains campus cohorts observed during spring academic terms. Results may not generalize to clinical, community, older, or seasonally different populations. Shared academic calendars may contribute to behavioral continuity.

Identity. Cross-year identity is incomplete. The report cannot establish a stable person-level trait across years. Excluding repeat candidates is conservative but does not reconstruct the missing identity clusters.

Released aggregates. Models use RAPIDS aggregate features rather than raw sensor streams. Results depend on feature extraction, missingness, platform, target definition, issue time, and model class.

Time scale. Early and late windows span weeks, not years or major life changes. Stability over these windows is not long-term reliability.

Internal gates. The 0.30 and 0.10 thresholds were recorded within the project and are not universal standards. Their repository record is not independently timestamped before analysis.

Entropy proxies. Marginal quintile entropy and short discretized sequence metrics do not estimate a theoretical predictability ceiling. Their estimates are sensitive to window length and discretization.

Pooled comparator. Relative history benefit depends on both a pooled history model and a pooled calendar model. Normalized calendar MAE varied modestly across cohorts. The score is not comparator-free.

Post-hoc development. The profile, feature-family, sequence, typology, and distress analyses were developed after earlier results. Their thresholds organize the exploration but do not provide external confirmation.

Uncertainty. Aggregate behavior means do not have one common confidence interval. Some secondary profile tests have only 500 permutations. H3 had limited power for small corrected associations.

Chapter 7

Conclusion and Future Work

7.1 Conclusions

This project gives four direct conclusions.

1. **Psychological-label validity is the primary result.** In the tested GLOBEM split, prior depression-label history achieved AUROC 0.844. Adding current-day or historical passive features produced 0.840 and 0.841. The passive features did not improve on label-only. Repeated psychological columns also showed substantial overlap.
2. **Direct behavior is prospectively forecastable.** Pooled history-based models achieved mean tomorrow $R^2 = 0.272$ and AUROC 0.788 across 16 direct targets. This is measurement feasibility for observed behavior.
3. **Scalar forecastability is not a validated new construct.** Raw forecastability correlated -0.912 with behavioral standard deviation. Its adjusted split-half stability was $r = 0.230$, below the internally recorded 0.30 gate. The scalar claim failed.
4. **A continuous profile remains only as a post-hoc lead.** The pattern of domain-specific pooled-model history benefit survived the tested controls. Cross-domain coupling and discrete types were unsupported. The profile is not an idiographic model or a stable person-level phenotype.

7.2 Recommended replication study

The next study should test the continuous profile in an independent dataset with longer trajectories and recoverable person identifiers. Its contract should be timestamped before outcome analysis. The study should compare:

1. pooled history models;
2. hierarchical or mixed-effects models; and
3. separately fitted person-specific models.

The primary question should be whether the same domain profile remains within the same person across independent periods and model classes. Variability, training-window moments, missingness, weekday structure, and preselected sequence metrics should be controls from the start.

7.3 Requirements before personalization

The stopped feasibility audit should be repeated only when the candidate dataset meets its frozen per-cohort and total eligibility requirements. A revised contract may use fewer target families, but that change must be justified before inspecting model outcomes. Feasibility and effect estimation must remain separate phases.

7.4 Final project statement

The project does not support an autonomous passive detector of a named psychological state, a scalar routine phenotype, or discrete routine types. It supports a clear measurement rule: compare passive models with label history, forecast direct behavior prospectively, and audit prediction-derived scores against simpler behavioral constructs. That rule is the defensible contribution of the work.

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Appendix A

Reproducibility and Source Map

Table A.1: Compact work-package source map. “Runnable” means a runner and checked output exist; restricted local GLOBEM inputs are still required where stated.

WP / status	Contract or definition; runner	Checked output	Decision represented
WP1 Runnable	experiments/README.md; experiments/run_label_baseli ne_control.py	results/label_baseline_contr ol/summary.json; metrics.csv	Label-history boundary
WP1 Runnable	experiments/README.md; experiments/run_construct_fr agility_audit.py	results/construct_fragility_ audit/summary.json; same_day _construct_correlations.csv	Construct-overlap boundary
WP2 Runnable	experiments/README.md; experiments/run_behavioural_ weather_forecast.py	results/behavioural_weather_ forecast/summary.json; regression_summary_r2.csv; classification_summary_auroc .csv	Prospective behavior result
WP3 Runnable	experiments/README.md; experiments/run_forecastabil ity_phenotype.py	results/forecastability/summ ary.json	Original scalar/profile test
WP3 Runnable	experiments/README.md; experiments/run_referee_robu stness_checks.py	results/referee_robustness/s ummary.json; stability_robustness.csv; profile_robustness.csv	Construct correction and refits
WP4 Runnable	experiments/README.md; experiments/run_structure_me chanism_followup.py; experiments/run_profile_typo logy_followup.py	results/structure_followup/s ummary.json; typology_summary.json	Post-hoc profile, mechanism, and typology
Secondary Runnable	experiments/README.md; experiments/run_common_distr ess_forecastability.py	results/distress_sensitivity /summary.json	Post-hoc distress sensitivity
WP5 Runnable audit	experiments/personalization/ contract.md; experiments/pe rsonalization/run.py	experiments/personalization/ result.json; experiments/pe rsonalization/review.md	Feasibility stop; no prediction model

A.1 Compact robustness record

Excluding all repeat-candidate trajectories retained 232 core trajectories and a raw fixed-model correlation of $r = 0.417$ with 95% interval $[0.261, 0.562]$. This shows that known repeats did not dominate the raw association; it does not validate the scalar or reconstruct unique identities. The sequence controls were based on short discretized records (median 68 observed days) and remain descriptive. H3 was inconclusive because its approximate corrected 80% minimum detectable $|r|$ was 0.172. The post-hoc distress estimates were heterogeneous across cohorts and do not support distress prediction or within-person change detection.

A.2 Canonical build and analysis command

To validate the public-safe aggregate evidence, run the tests, regenerate all six report figures, and compile the PDF, run from the repository root:

```
python scripts/reproduce_report.py verify
```

After obtaining GLOBEM v1.1 under its access terms and placing it under the local path documented in `DATA_ACCESS.md`, recompute the full analysis with:

```
python scripts/reproduce_report.py full
```

The full command rebuilds the report-relevant analyses before running the same verification. GLOBEM source data are required locally and are not redistributed. The exact environment is pinned in `requirements-lock.txt`; the data boundary and expected local layout are documented in `DATA_ACCESS.md`. The full headline-claim and display map is maintained in `report/CLAIM_SOURCE_MAP.md` rather than duplicated in the PDF.